

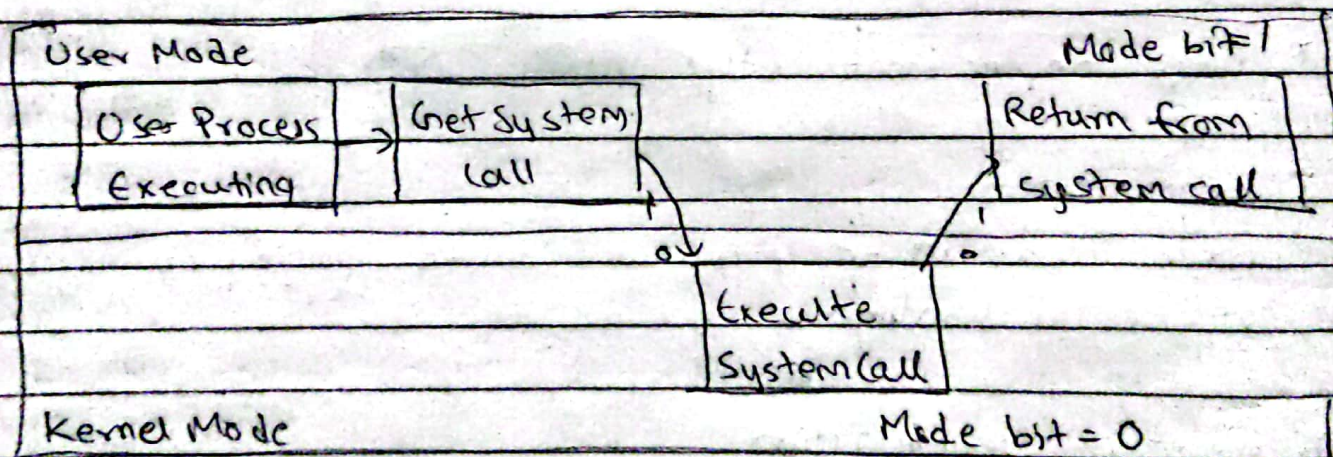
Chapter 2

Structure of Operating System.



* User Mode vs Kernel Mode

- ↳ As an user when we turn on the laptop & use some application there, then by default we are in the user mode.
- ↳ But all core functionalities of OS, all major drivers, all works & executes in the kernel mode.
- ↳ Meanwhile, the processor keeps switching between user mode & kernel mode.
- ↳ Eg: If you want to write a program to do $2+2=4$ then you don't need kernel.
But if you want output 4 to get printed on monitor then we'll give printf, here monitor is in OS's hand so system call will be generated & it will print value on monitor by switching betⁿ user & kernel mode.



- ↳ All the hardware are in control of kernel or OS, so user can't access the hardware directly & it needs kernel.
- ↳ So, if user is using any application or API or writing any program, then we write in user mode but if it wants to access any functionality of OS then it has to go in kernel mode.

* System call

↳ is a programmatic way through which we shift from user mode to kernel mode.

Types:

• File Related System Calls:

↳ Means where we want to access any file.

↳ `Open()`, `Read()`, `Write()`, `Close()`, `CreateFile`, etc.

• Device Related System Calls:

↳ if I want to access any device or hardware whether it is harddisk, printer or monitor which a user can never access directly but has to take privilege from OS.

↳ `Read`, `Write`, `Reposition`, `ioctl`, `fcntl`

• Information Related:

↳ regarding processes. ~~etc~~ \$

↳ `getPid`, `attributes`, `get system time and data`

• Process Control Related:

↳ We write or save code in secondary memory in harddisk to save it permanently.

↳ But when we have to load them in main memory or execute, then we use process control system call.

↳ `Load`, `Execute`, `abort`, `fork`, `Wait`, `Signal`, `Allocate`, etc.

• Communication:

↳ We do inter process communication

↳ `Pipe()`, `Create/delete connections`, `shmget` etc.
(to get value of shared memory)

* Kernel

- ↳ Whenever a system starts, the kernel is the first program that is loaded after the boot loader because the kernel has to handle rest of the thing of system for OS.
- ↳ The kernel remains in the memory until the OS is shut-down.

↳ ~~Other~~

* Functions of Kernel:

- 1) Access Computer Resource (like CPU, I/O devices)
- 2) Resource Management
- 3) Memory Management
- 4) Device Management
- 5) Interrupt Handling

* Types of Kernel:

1) Monolithic kernels:

- ↳ where user services & kernel services are implemented in same memory space i.e. different memory is not used.
- ↳ So the execution of process will be faster.

2) Microkernel

- ↳ different spaces.
- ↳ reduces speed of execution.

3) Hybrid kernel

- ↳ combined aspect of monolithic & microkernel.
- ↳ makes use of speed of monolithic & modularity of